Hung-Ting Su

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Reliable Multimodal AI • Robot Learning • LLM / MLLM Reasoning • Evaluation

Summary

- Researcher-engineer working on reliable multimodal AI, spanning AI safety, robot learning, reasoning, and evaluation.
- Recent industry work at Delta spanned robot-learning infrastructure, tactile and visual policy learning, VLA-style manipulation, emerging world-model directions, rollout evaluation, and failure-driven improvement for real-world manipulation.
- Academic research spans trustworthy LLM/MLLM reasoning, long-form video understanding, and multimodal evaluation, developed through work at NTU and Columbia and now extended at NUS to harmful video content detection and adversarial robustness.

Experience

Research Fellow Sep 2026–Present
National University of Singapore Singapore

- Research harmful video content detection and the robustness of multimodal language models to adversarial attacks, including prompt injection, and investigate defenses.

Senior Associate Researcher 2025–Aug 2026
Delta Robotics Innovation Center, Delta Electronics Taiwan

- Built robot-learning infrastructure and embodied-AI prototypes that connect multimodal data, VLA-style policies, emerging world-model directions, evaluation loops, and real-world deployment.
- Developed tactile and visual policy-learning workflows for contact-rich manipulation, including failure analysis and targeted iteration from rollout outcomes.
- Collaborated across robotics, sensing, and AI components to move research systems toward robust end-to-end operation.

Postdoctoral Researcher 2023–2025
National Taiwan University (NTU) Taiwan

- Led and mentored projects across embodied AI, trustworthy LLM/MLLM reasoning, long-form video understanding, and multimodal evaluation.
- Built research directions that connect navigation, planning, movie reasoning, automatic verification, and long-horizon evidence-seeking behavior.
- Led projects resulting in CoRL 2024, EMNLP 2024, and ACL 2026, and mentored student-led work accepted at NeurIPS 2024, ICCV 2025, EMNLP 2025, AAAI 2026, and ACL 2026.

Visiting Researcher 2022–2023
Columbia University, hosted by Prof. Shih-Fu Chang New York, NY

- Developed early zero-shot causal video reasoning frameworks and helped establish a cross-institution collaboration between NTU and Columbia.
- This collaboration led to an EMNLP 2024 publication and strengthened later work on trustworthy multimodal reasoning.

Selected Publications

Full publication history: Google Scholar.

- **VLN-NF: Feasibility-Aware Vision-and-Language Navigation with False-Premise Instructions** *(ACL 2026)* [paper] [project page] — Evidence-seeking embodied navigation with explicit abstention under false-premise or infeasible instructions. **Role: Lead**
- **Unveiling Narrative Reasoning Limits of Large Language Models with Trope in Movie Synopses** *(EMNLP 2024)* [paper] [code] — Analyzes narrative reasoning limits and chain-of-thought hallucinations in LLMs using movie tropes. **Role: Lead**
- **Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation** *(CoRL 2024)* [paper] [code] — Zero-shot visual map refinement and semantic replanning for more robust embodied navigation. **Role: Lead**
- **AED: Adaptable Error Detection for Few-shot Imitation Policy** *(NeurIPS 2024)* [paper] — Policy monitoring for few-shot imitation learning with deployment-oriented error detection. **Role: Mentor**
- **MovieCORE: COgnitive REasoning in Movies** *(EMNLP 2025)* [paper] [project page] — Long-form movie benchmark for cognitive and social reasoning with multimodal models. **Role: Mentor**
- **HERMES: temporal-coHERent long-forM understanding with Episodes and Semantics** *(ICCV 2025)* [paper] [project page] — Improves long-form multimodal understanding through temporal coherence and episodic structure. **Role: Mentor**
- **Affordance-Guided Coarse-to-Fine Exploration for Base Placement in Open-Vocabulary Mobile Manipulation** *(AAAI 2026)* [paper] — Training-free VLM-guided planning for mobile-manipulation base placement in open-vocabulary settings. **Role: Mentor**
- **ADAPT: Benchmarking Commonsense Planning under Unspecified Affordance Constraints** *(ACL 2026)* [paper] — Benchmarking and evaluating planning under underspecified affordance and feasibility constraints. **Role: Mentor**

Education

PhD in Computer Science 2023
National Taiwan University (NTU), Advisor: Prof. Winston H. Hsu Taiwan

- Dissertation: *Multi-modal Video Comprehension and Beyond*
- Visiting researcher at Columbia University, hosted by Prof. Shih-Fu Chang.

MS in Computer Science 2017
National Taiwan University Taiwan

BS in Computer Science 2013
National Tsing Hua University Taiwan

Skills

- **Research Areas:** robot learning, embodied AI, multimodal reasoning, long-form video understanding, and evaluation under uncertainty.
- **Core Methods:** imitation learning, VLA-style policies, world-model-oriented planning, failure analysis, benchmark construction, data generation, and automatic verification.
- **Technical Stack:** Python, PyTorch, HuggingFace Transformers, ROS2, Linux, Docker, Git, RGB-D/3D perception, and tactile/contact sensing.
- **Collaboration:** student mentoring, cross-institution research coordination, technical writing, and publication-driven project leadership.

Academic Services

- Area Chair: NAACL/COLING 2027 (ARR 10/2026); ACL 2026 (ARR 01/2026); EACL 2026 (ARR 10/2025).
- Reviewer: ICML, NeurIPS, CVPR, ICCV, NAACL, ACL, EMNLP, CoRL, ICRA, IROS, AAAI, ACM Multimedia, TCSVT.

References

References are available upon request.